

For all questions below, you should write sample code, which can be ran directly on online compilers such as https://www.onlinegdb.com/online_java_compiler (if possible) to explain your answer

1. Write up Example code to demonstrate the three fundamental concepts of OOP.
 1. Encapsulation;
 2. Polymorphism;
 3. Inheritance;
2. What is **wrapper data type classes** (e.g. Integer, Double) in Java and Why we need wrapper class?
3. What is the difference between **HashMap** and **HashTable**?
4. What is **String pool** in Java and why we need String pool? Explain String immutability.
5. Explain **garbage collection**? Explain types of garbage collection.
6. What are **access modifiers** and their scopes in Java?
7. Explain **final** key word? (File, Method, Class)
8. Explain **static** keyword? (File, Method, Class). When do we usually use it?
9. What is the differences between **overriding** and **overloading**?
10. Explain how Java defines a **method signature**, and how it helps on overloading and overriding.
11. What is the differences between **super** and **this**?
12. Explain how `equals` and `hashCode` work.
13. What is the Java **load sequence**?
14. What is **Polymorphism** ? And how Java implements it ?
15. What is **Encapsulation** ? How Java implements it? And why we need encapsulation?
16. Compare **interface** and **abstract class** with use cases.